

$$-\log(I/I_0) = \int_L \mu(r) \, ds$$

$$y = Ax + \eta$$

$$dz = f(z, \tau) d\tau + g(\tau) dv$$

$$dz = \left[f(z, \tau) - g(\tau)^2 \nabla_z \log p_\tau(z) \right] d\tau + g(\tau) d\bar{w}$$

$$z_{\sigma} = x + \sigma \epsilon, \quad \epsilon \sim \mathcal{N}(0, I)$$

$$s_{\phi}(z, \sigma) = \frac{D_{\phi}(z, \sigma) - z}{\sigma^2}$$

$$\hat{x}_0 \stackrel{=}{=} D_\phi(z, \sigma), \quad \hat{y} \stackrel{=}{=} Ax_0$$

$$s_{\phi,c}(G_c z, \rho_c, \sigma) = \frac{D_{\phi}(G_c z, \rho_c, \sigma) - G_c z}{\sigma^2}$$

$$\boldsymbol{s}_{\text{PaDIS}}(z, \sigma) \approx \sum_{r=1}^{(k+1)^2} G_{i,j,r}^{\text{T}} \boldsymbol{s}_{\phi,i,j,r}(G_{i,j,r} z, \boldsymbol{\rho}_{i,j,r}, \sigma)$$

$$\hat{x}_0^k = D_\phi(z^k, \sigma_k)$$

$$\tilde{z}^{k+1} = z^k - \frac{S^k}{2} \nabla_{z^k} \left\| y - Ax_0^k \right\|_2^2$$

$$z^{k+1} = \tilde{z}^{k+1} + \frac{a_k}{2} s_\phi(\tilde{z}^{k+1}, \sigma_k) + \sqrt{a_k} \epsilon^k$$

$$\hat{x} = \arg \min_x \frac{1}{2} ||y - Ax||_2^2 + \lambda R(x)$$

$$\min_x \max_{p, \|q_i\|_2 \leq \lambda} \langle Ax - y, p \rangle + \langle Dx, q \rangle - \frac{1}{2} \|p\|_2^2$$

$$(A^{\top}A + \eta I)x^{k+1} = A^{\top}y + \eta(v^k - u^k)$$

$$v^{k+1} \equiv D_{\sigma}(x^{k+1} + u^k), \quad u^{k+1} \equiv u^k + x^{k+1} - v^{k+1}$$

$$\hat{x}_y^k = By + (I - BA)\hat{x}_0^k$$

$$z_{\text{data}}^k = z^k - \frac{\zeta}{\|r^k\|_2} \nabla_{z^k} \|r^k\|_2^2$$

$$z_{\text{data}}^k = z^k - \zeta_{\text{CDPS}} \nabla \|z^k\| \|r^k\|/2$$

$$z_{\text{data}}^k = z^k - \frac{\zeta}{L_F} \nabla_{z^k} \frac{1}{2} \|r^k\|^2$$

$$L_F = \|F\|_2^2$$

$$\text{MAE} = \frac{1}{N} \|\hat{x} - x\|_1$$

$$\text{PSNR} = 10 \log_{10} \left(\frac{1}{\text{MSE}} \right)$$

$$\text{SSIM}(x, \hat{x}) = \frac{(2\mu_x\mu_{\hat{x}} + C_1)(2\sigma_{x\hat{x}} + C_2)}{(\mu_x^2 + \mu_{\hat{x}}^2 + C_1)(\sigma_x^2 + \sigma_{\hat{x}}^2 + C_2)}$$

$$\text{RSR} = \frac{\|A\hat{x} - y\|_2}{\|y\|_2}$$

$$c_{\text{DPS}} = 0.0405, \quad c_{\text{adj}} = 0.1022$$

$$g(\tau) = \sqrt{\frac{d[\sigma^2(\tau)]}{dt}}, \quad f(z, \tau) = 0$$